SE3082

AWS EC2 Lab - Screenshots and Calculations

Lab 08

Exercise 0: In-class activities (HelloMPI and message passing)

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On instance with 2 vCPUs (t3.micro)

Nano - creating the HelloMPI code

```
  GNU nano 6.2                                                HelloMPI.cc
#include <cstdio>
#include <cstdlib>
#include <mpi.h>
int main(void)
{
    int rank;
    MPI_Init(NULL, NULL);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    char name[30];
    int len;
    MPI_Get_processor_name( name, &len );
    if (rank == 0)
      printf("Hello World! From rank %d machine %s\n", rank, name);
    else
      printf("Just a normal process From rank %d machine %s\n", rank, name);
    MPI_Finalize();
}
```

Outputs from running the HelloMPI code

Running with 2 processes

```
[ec2-user@ip-172-31-17-201 ~]$ mpirun --oversubscribe -np 2 ./HelloMPI
Hello World! From rank 0 machine ip-172-31-17-201.ec2.internal
Just a normal process From rank 1 machine ip-172-31-17-201.ec2.internal
```

Running with 4 processes

```
[ec2-user@ip-172-31-17-201 ~]$ mpirun --oversubscribe -np 4 ./HelloMPI
Just a normal process From rank 3 machine ip-172-31-17-201.ec2.internal
Just a normal process From rank 1 machine ip-172-31-17-201.ec2.internal
Just a normal process From rank 2 machine ip-172-31-17-201.ec2.internal
Hello World! From rank 0 machine ip-172-31-17-201.ec2.internal
```

Nano - creating the messages1 code

```
 GNU nano 6.2                                          messages1.cc
#include <cstdio>
#include <cstdlib>
#include <mpi.h>
int main(void)
{
    int rank;
    MPI_Status status;
    MPI_Init(NULL, NULL);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    char name[30];
    int len;
    MPI_Get_processor_name( name, &len );
    int x, y;
    x = 30;
    if (rank == 1) {
      printf("SEnding message to computer 3 from computer 1\n");
      MPI_Ssend(&x, 1, MPI_INT, 3, 0, MPI_COMM_WORLD);
    }
    else if (rank == 3) {
      MPI_Recv(&y, 1, MPI_INT, 1, 0, MPI_COMM_WORLD, &status);
      printf("in computer 3 the value of y is %d\n", y);
    }
    else
      printf("Just a normal process From rank %d machine %s\n", rank, name);
    MPI_Finalize();
}
                                                  [ Read 26 lines ]
```

Output from running the messages1 code (4 processes)

```
[ec2-user@ip-172-31-17-201 ~]$ mpic++ messages1.cc -o messages1
[ec2-user@ip-172-31-17-201 ~]$ mpirun --oversubscribe -np 4 ./messages1
Just a normal process From rank 0 machine ip-172-31-17-201.ec2.internal
SEnding message to computer 3 from computer 1
in computer 3 the value of y is 30
Just a normal process From rank 2 machine ip-172-31-17-201.ec2.internal
```

Nano - creating the message2 code

```
  GNU nano 6.2                                                  message2.cc
#include <cstdio>
#include <cstdlib>
#include <mpi.h>
int main(void)
{
    int rank;
    MPI_Status status;
    MPI_Init(NULL, NULL);
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    char name[30];
    int len;
    MPI_Get_processor_name( name, &len );
    int x[10], y[10];
    if (rank == 1) {
      for (int r =0;r <10; r++)
         x[r] = 10*r;

      printf("SEnding message to computer 3 from computer 1\n");
      MPI_Ssend(x, 10, MPI_INT, 3, 0, MPI_COMM_WORLD);
    }
    else if (rank == 3) {
      MPI_Recv(y, 10, MPI_INT, 1, 0, MPI_COMM_WORLD, &status);
      printf("in computer 3 the value of y is printed\n");
      for (int r=0;r<10;r++)
         printf(" %d ",y[r]);
    }
    else
      printf("Just a normal process From rank %d machine %s\n", rank, name);
    MPI_Finalize();
}
```

Output from running the message2 code (4 processes)

```
[ec2-user@ip-172-31-17-201 ~]$ mpirun --oversubscribe -np 4 ./message2
SEnding message to computer 3 from computer 1
Just a normal process From rank 2 machine ip-172-31-17-201.ec2.internal
Just a normal process From rank 0 machine ip-172-31-17-201.ec2.internal
in computer 3 the value of y is printed
 0  10  20  30  40  50  60  70  80  90 [ec2-user@ip-172-31-17-201 ~]$
```